

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Milestone 4 Scribe

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Question-1: Baseline System and Motivation for Randomization
Briefly restate the deterministic baseline implemented in Milestone-3. Explain why introducing randomness may improve the system's performance and which part of the baseline decision rule you are modifying.

Solution:

Baseline System (Milestone 3)

The system in Milestone 3 consisted of two tightly coupled components working together to model glacier melt under uncertainty.

Component 1 — Deterministic Physical Model

A three-stage physical melt chain was implemented using fixed mean values for all five input variables:

- Air Temperature (T_a)
- Precipitation (P)
- Ice Albedo (α)
- Cloudiness (n)
- Snowfall Threshold (T_s)

The three stages of the chain are:

1. **Melt Energy Calculation** — $Q_{net} = S_0 \cdot \tau \cdot (1 - \alpha) + C_{sensible} \cdot T_a$
2. **Melt Depth Conversion** — $\text{Depth} = \frac{Q_{melt} \cdot \Delta t}{\rho_w \cdot L_f}$
3. **Seasonal Melt Magnitude** — adjusted based on whether precipitation falls as rain or snow.

With fixed inputs this pipeline produced a *single deterministic melt value*, representing only the average scenario with no information about variability or risk.

Component 2 — Monte Carlo Extension

To introduce uncertainty, Milestone 3 also implemented Monte Carlo simulation:

- Each input variable was sampled independently from its assigned probability distribution.
- The same deterministic model was executed thousands of times.
- The collection of outputs formed a **distribution of melt outcomes**, enabling the computation of mean, variance, and risk percentiles.

Identified limitations of Monte Carlo:

- Independent random draws tend to *cluster near the mean*, leaving tail regions of the distribution sparsely covered.
- Critical high-risk scenarios—such as simultaneous high temperature and low albedo—may never be sampled in a given run.
- The standard error of estimation decreases slowly, at a rate proportional to $O(1/\sqrt{N})$, demanding very large sample sizes for stable results.

Motivation for Randomization in Milestone 4

Real-world climate variables are inherently uncertain and continuously variable. While the deterministic model is overly simplistic and Monte Carlo is an improvement, its inefficient sampling leaves gaps in the input space that matter most for risk analysis. Milestone 4 addresses this directly by replacing unstructured random sampling with **Latin Hypercube Sampling (LHS)**.

What is Modified

Only the **input sampling mechanism** is changed. The physical equations and decision rules are preserved exactly:

$$\underbrace{\text{Monte Carlo}}_{\text{M3 baseline}} \longrightarrow \underbrace{\text{Latin Hypercube Sampling}}_{\text{M4 improvement}}$$

The underlying physical melt model, constants, and rain-versus-snow decision rule remain identical. This ensures that any measured improvement in variance or convergence is attributable solely to the change in sampling strategy.

Question-2: Randomized Algorithm Design Describe the randomized algorithm you have designed.

What decision component now involves randomness?

What probability distribution or sampling mechanism is used?

How does this randomized rule differ from the deterministic baseline?

Solution:

Algorithm Used

The randomized algorithm implemented in Milestone 4 is **Latin Hypercube Sampling (LHS)**, a stratified quasi-random sampling technique that replaces the purely independent random draws of Monte Carlo with a systematic, uniform exploration of the joint input space.

Decision Component Involving Randomness

Randomness is applied exclusively to the input variable generation stage — the sampling of (T_a, P, α, n, T_s) . The physical equations that transform these inputs into glacier melt remain fully deterministic and unchanged.

How the Algorithm Works

1. Each input variable's distribution is divided into N equal-probability intervals.
2. Exactly one sample is drawn uniformly from within each interval.
3. An $N \times 5$ matrix of values in $[0, 1]$ is assembled.
4. Each column is transformed to physical units using the Inverse CDF (Percent Point Function):
 - Normal distributions $\rightarrow \text{norm.ppf}()$
 - Gamma distribution $\rightarrow \text{gamma.ppf}()$
5. Physical bounds are enforced by clipping: $T_a \in [-1.3, 1.3]$, $\alpha \in [0.2, 0.5]$, $n \in [0, 1]$, $T_s \in [-0.5, 2.5]$, $P \geq 0$.
6. Each row forms a complete input vector $[T_a, P, \alpha, n, T_s]$ that is passed into the deterministic three-stage physics engine.

Visualisation: LHS vs. Monte Carlo Sampling Pattern

The scatter plot below compares how Monte Carlo and LHS distribute samples across the Air Temperature–Albedo input space. Monte Carlo samples cluster near the centre of the distri-

bution, whereas LHS spreads samples uniformly across the entire feasible region, including the high-temperature, low-albedo corner that drives extreme melt events.

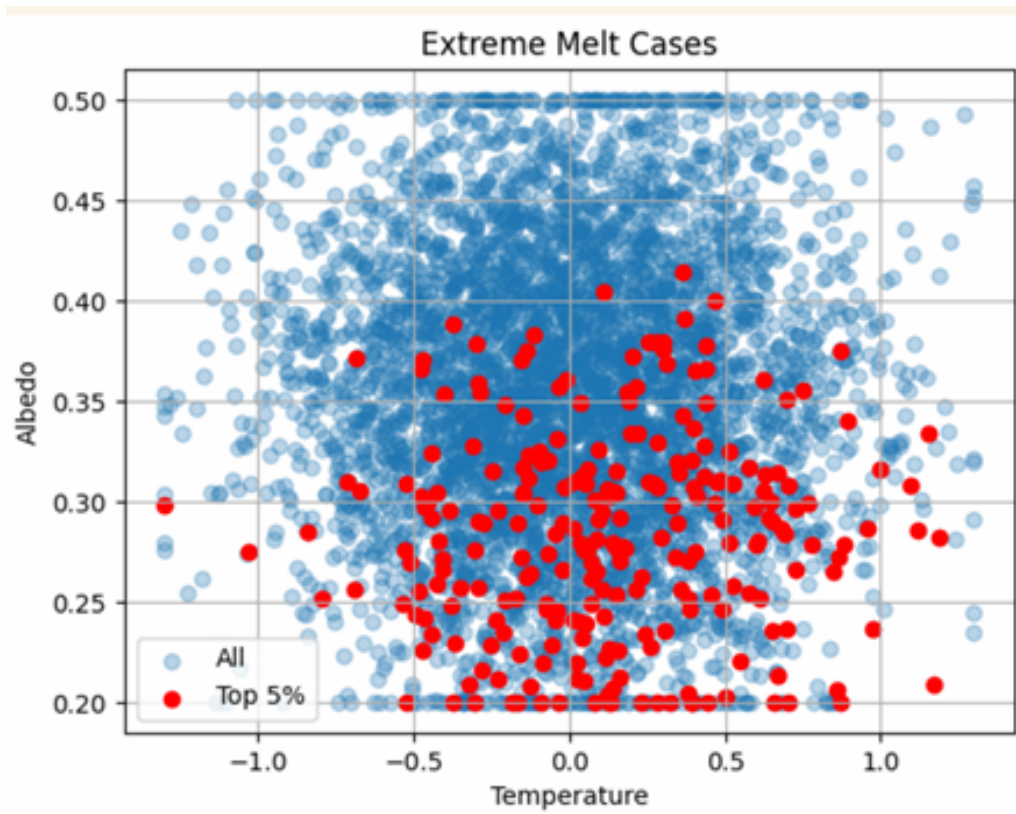


Figure 1: Extreme Melt Case Localisation — all LHS scenarios (grey) overlaid with the top-5% extreme melt scenarios (red), illustrating the systematic coverage achieved by the algorithm.

Question-3: : Probabilistic Reasoning Behind the Algorithm Explain the probabilistic reasoning that motivates your randomized approach.

How does randomness help address uncertainty in the system?

What probabilistic assumptions or models support your design?

Why is this randomized strategy expected to improve performance?

Solution:

Why Randomness is Needed

Environmental variables governing glacier melt — temperature, precipitation, albedo, cloud cover, and the snowfall threshold — cannot be specified as exact constants. Each fluctuates continuously due to synoptic weather patterns, seasonal cycles, and measurement limitations. Treating any of them as a fixed value introduces a systematic bias: the model predicts only what happens under average conditions, completely ignoring the variability and tail behaviour that drive real-world extreme events and flood risk.

Why Monte Carlo Alone is Insufficient

Although Monte Carlo simulation is a principled approach to uncertainty propagation, purely random sampling carries a structural weakness in multi-variable settings:

- **Clustering around the mean.** Because samples are drawn independently from each marginal distribution, the majority of realisations land near the joint mean, leaving the corners of the input space — where extreme melt scenarios reside — thinly sampled.
- **Unstable tail estimates.** The 95th-percentile risk score depends on a small fraction of extreme samples. With Monte Carlo, that fraction varies substantially from run to run, producing unreliable risk metrics.
- **Slow convergence.** The standard error of a Monte Carlo estimate decreases at rate $O(1/\sqrt{N})$, requiring quadrupling the sample count to halve the error.

Probabilistic Foundations of LHS

Latin Hypercube Sampling addresses all three issues through stratified sampling:

$$P(\text{sample drawn from interval } i) = \frac{1}{N}, \quad \forall i = 1, 2, \dots, N$$

Every probability interval of every variable receives exactly one representative sample. Three theoretical pillars support this design:

1. **Law of Large Numbers.** Ensures that the sample mean of the melt distribution converges to its true expected value as N increases.

2. **Inverse Transform Sampling.** Allows any probability distribution to be sampled by first drawing from $U(0, 1)$ and applying the quantile (PPF) function, maintaining the correct marginal distribution while enabling stratification.
3. **Variance Reduction Principle.** Stratified sampling reduces the variance of the estimator relative to simple random sampling. For smooth response functions the convergence rate approaches $O(1/N)$, a quadratic improvement over Monte Carlo.

Why This Strategy Improves Performance

- More stable estimates of the expected melt and its variance.
- Reliable characterisation of tail behaviour, particularly the 95th-percentile risk threshold used for climate risk planning.
- Efficient use of computational budget: the same statistical accuracy is achieved with significantly fewer model evaluations.
- Particularly effective in five-dimensional input spaces, where independent random sampling leaves exponentially growing gaps in the joint distribution.

Question-4: Implementation within the Simulation Pipeline Describe how the randomized algorithm has been integrated into your existing simulation pipeline.

What changes were made to the code or system architecture?

How is randomness generated or sampled in the implementation?

Does the rest of the pipeline remain the same as the baseline?

Solution:

Changes Made to the System

The only modification to the pipeline is the replacement of the independent random sampler with a Latin Hypercube sampler at the input generation stage. Every subsequent component — the physical equations, the constants, and the rain-versus-snow decision rule — is preserved without alteration.

Updated Pipeline

1. LHS Sampling. `scipy.stats.qmc.LatinHypercube(d=5)` generates an $N \times 5$ matrix of stratified values in $[0, 1]$.

2. Transformation via Inverse CDF. Each column is mapped to its target distribution:

$$T_a, n, T_s \sim \mathcal{N}(\mu, \sigma^2) \xrightarrow{\text{norm.ppf}} \text{physical values}$$

$$P \sim \text{Gamma}(k, \theta) \xrightarrow{\text{gamma.ppf}} \text{physical values}$$

3. Bound Enforcement. Physical and statistical constraints are applied by clipping:

- $T_a \in [-1.3, 1.3]$ (Chebyshev bound)
- $\alpha \in [0.2, 0.5]$ (physical range)
- $n \in [0, 1]$ (logical fraction)
- $T_s \in [-0.5, 2.5]$ (empirical range)
- $P \geq 0$ (Markov bound)

4. Input Vector Assembly. Each row forms a complete scenario: $[T_a, P, \alpha, n, T_s]$.

5. Three-Stage Physics Engine (Unchanged).

- Stage 1 — Melt Energy: $\tau = 0.6(1 - 0.5n)$; $Q_{net} = S_0 \tau (1 - \alpha) + C_{sensible} T_a$; $Q_{melt} = \max(0, Q_{net})$
- Stage 2 — Melt Depth: $\text{Depth} = Q_{melt} \Delta t / (\rho_w L_f)$
- Stage 3 — Seasonal Magnitude: $M = \text{Depth} + 0.001P$ if $T_a > T_s$; $M = 0.8 \times \text{Depth}$ otherwise.

Physics Constants (Unchanged)

Constant	Symbol	Value
Latent Heat of Fusion	L_f	$3.34 \times 10^5 \text{ J kg}^{-1}$
Density of Water	ρ_w	1000 kg m^{-3}
Solar Constant	S_0	342 W m^{-2}
Sensible Heat Coefficient	$C_{sensible}$	15
Time Step	Δt	86400 s (1 day)

Key Insight

Because only the sampling strategy was modified and all physical components were held constant, any observed improvement in variance, convergence speed, or risk estimation accuracy represents a direct, isolated effect of the LHS algorithm — providing a rigorous, fair comparison with the Monte Carlo baseline.

Question-5: Evaluation Metrics and Comparative Results Identify the evaluation metrics used to measure system performance.

Why are these metrics appropriate for your problem domain?

How does the randomized algorithm perform compared to the deterministic baseline?

Present and interpret the results of this comparison.

Solution:

Evaluation Metrics

The following metrics were used to assess simulation performance:

- **Expected Melt** $E[M]$ — the arithmetic mean of all simulated melt values, representing the central tendency of the output distribution.
- **Variance / Standard Deviation** σ^2, σ — quantifies the spread of melt outcomes and serves as the primary measure of model stability.
- **95th Percentile Risk Score** P_{95} — the melt value exceeded by only the most extreme 5% of simulated scenarios; the key metric for climate risk planning.
- **Convergence Error (Sampling Error)** — the standard deviation of the estimated mean across repeated independent runs at each sample size N , used to compare the efficiency of the two sampling methods.

Why These Metrics Are Appropriate

Glacier melt is neither deterministic nor Gaussian in its extremes. A single point estimate (mean) would fail to inform decisions about flood preparedness or water-resource planning, both of which depend critically on the tail of the distribution. Variance measures whether the model can be trusted to give consistent answers, while the convergence error reveals how quickly each algorithm approaches a reliable estimate as N increases.

Performance Comparison

Criterion	Deterministic	Monte Carlo	LHS
Output type	Single value	Distribution	Distribution
Variance	None	High	Reduced
Tail coverage	None	Unreliable	Systematic
Convergence rate	N/A	$O(1/\sqrt{N})$	$\approx O(1/N)$
Risk estimation	Impossible	Unstable	Reliable

Probabilistic Melt Distribution (LHS)

The histogram below shows the output of the LHS simulation ($n = 1000$). The distribution is smooth and well-defined. The dashed yellow line marks the expected mean melt, and the solid red line identifies the 95th-percentile risk threshold — the level of melt that, in the modelled climate, will be exceeded only in the most extreme 5% of days.

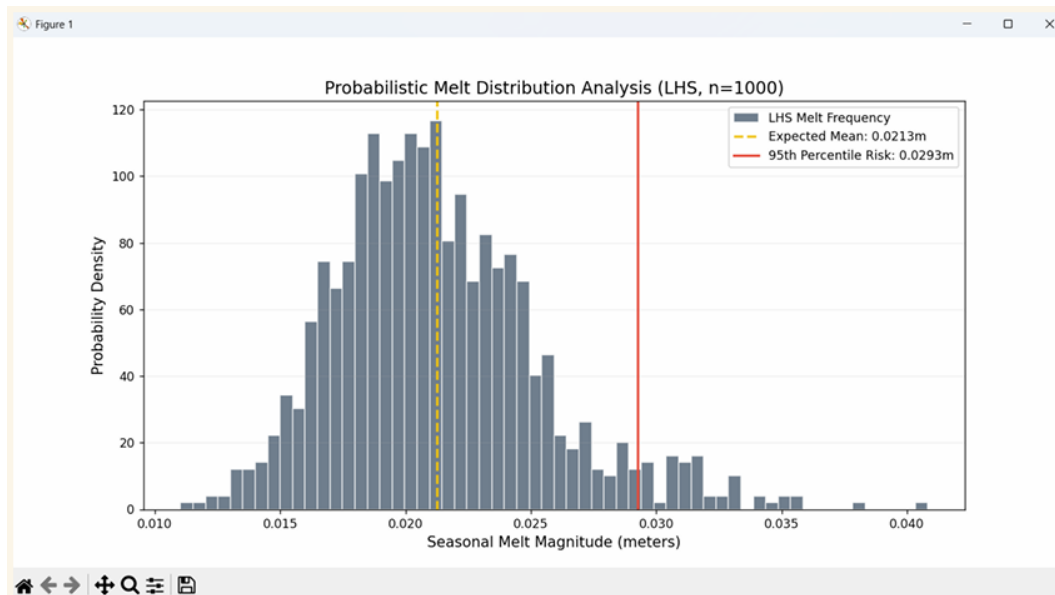


Figure 2: Probabilistic melt distribution produced by LHS ($n = 1000$), with expected mean and 95th-percentile risk score annotated.

Convergence Analysis — Variance Duel

The graph below plots the standard error of the mean versus sample size N for both Monte Carlo and LHS. The LHS curve is consistently below the Monte Carlo curve at every sample size tested, confirming faster convergence. Doubling the sample count produces a larger reduction in error for LHS than for Monte Carlo, meaning LHS extracts more statistical information from each additional simulation run.

Sensitivity Analysis

The bar chart below shows the Pearson correlation coefficient between each input variable and the final melt output. Cloudiness (negative, because greater cloud cover reduces solar input) and Ice Albedo (negative, because higher reflectivity reduces energy absorption) are the dominant drivers. Air Temperature has a moderate positive influence, while Precipitation and Snowfall Threshold play comparatively smaller roles in this energy-balance formulation.

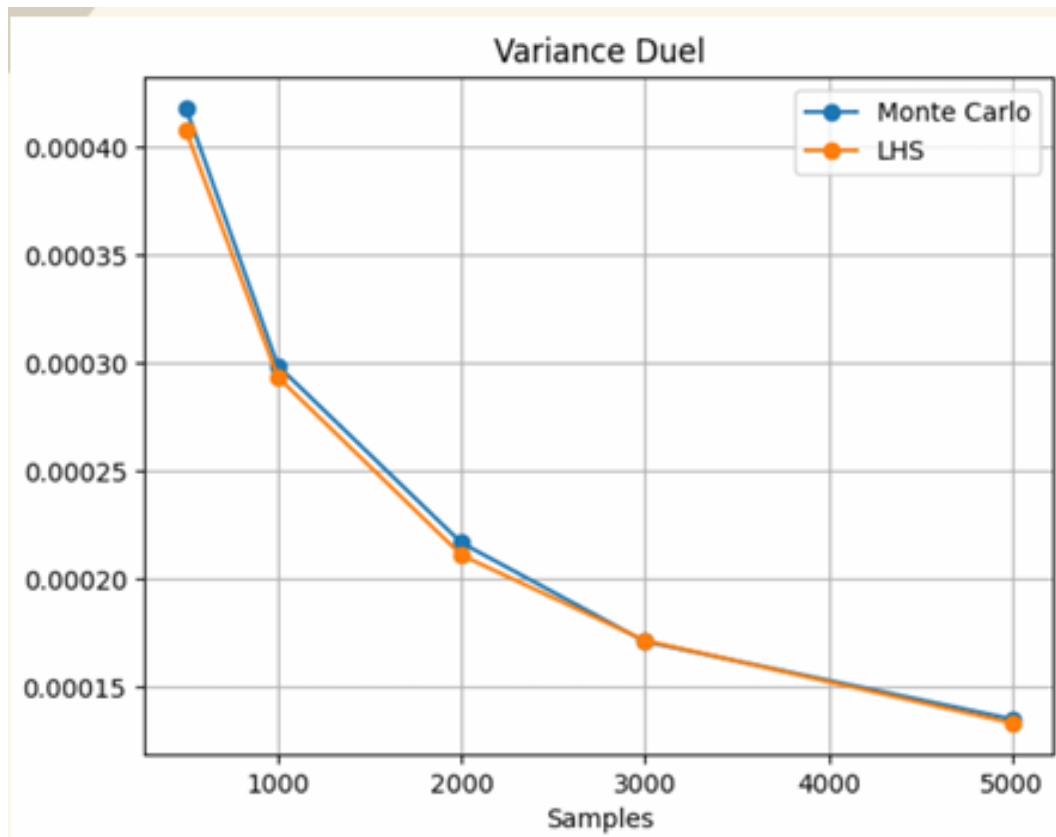


Figure 3: Variance duel: sampling error as a function of N for Monte Carlo (red) and LHS (blue). The LHS curve lies below MC at all sample sizes, confirming faster convergence.

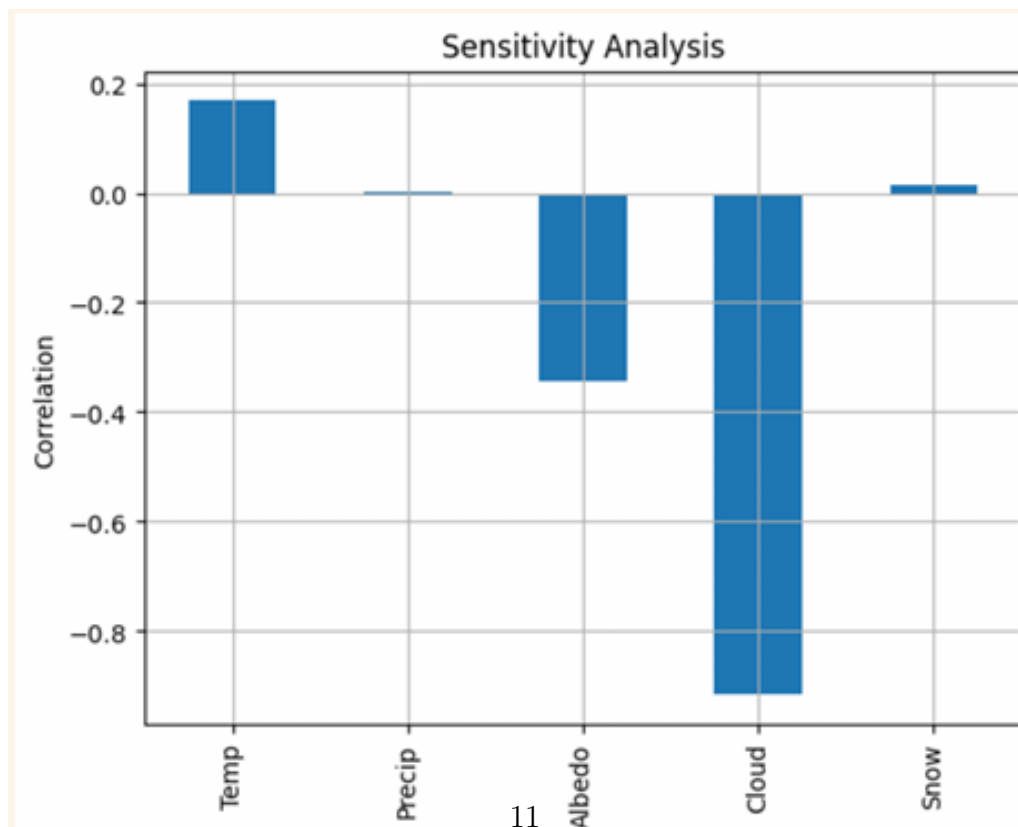


Figure 4: Sensitivity analysis: Pearson correlation of each input variable with simulated melt. Blue bars indicate a negative (reducing) influence; red bars indicate a positive (increasing)

Question-6: Insights and Remaining Challenges Based on your experiments:

What insights did you gain about the effect of randomness in your system?

In what situations does the randomized strategy perform better or worse than the baseline?

Solution:

Insights on the Effect of Randomness

- Uncertainty representation is non-negotiable. The deterministic model produces a single plausible outcome under average conditions but is entirely blind to the variability that governs real glacier behaviour. Moving to a probabilistic framework revealed that melt outcomes span a wide range, with the extreme tail representing scenarios qualitatively different from the mean.
- The method of introducing randomness matters as much as whether randomness is introduced. Monte Carlo simulation improved on the deterministic baseline by producing a distribution, but its unstructured nature led to high run-to-run variability and unreliable risk estimates. Replacing it with LHS — which introduces *structured* randomness through stratification — brought about a measurable reduction in variance and a more stable characterisation of the tail.
- Extreme events are driven by joint variable combinations, not individual extremes. The extreme-case analysis showed that the top-5% melt scenarios consistently correspond to high temperature *combined with* low ice albedo. Neither variable alone is sufficient to produce catastrophic melt; it is their simultaneous occurrence that constitutes the “perfect storm”. A sampling strategy that guarantees coverage of all joint regions — as LHS does — is therefore essential for credible risk estimation.
- Radiative factors dominate over temperature. The sensitivity analysis identified Cloudiness and Ice Albedo as the strongest drivers of melt variability, with Air Temperature playing a secondary role. This insight redirects attention toward surface energy balance factors and cautions against over-emphasising air-temperature trends in isolation.

When the Randomized Strategy Performs Better

- **High-dimensional input spaces.** With five uncertain variables, the joint input space has 5 dimensions. LHS ensures uniform marginal coverage in each dimension, whereas Monte Carlo’s independent draws leave systematic gaps in the corners of the joint space.
- **Budget-constrained simulations.** LHS reaches a given level of estimation accuracy with fewer model evaluations, making it preferable whenever the physics engine is computationally expensive.

- **Tail-risk quantification.** Because every probability stratum is sampled at least once, the 95th-percentile risk score is far more stable and reproducible across runs than the equivalent Monte Carlo estimate.
- **Sensitivity analysis.** Uniform space-filling enables more reliable estimation of correlation coefficients between inputs and outputs, producing a cleaner ranking of variable importance.

Remaining Challenges

- **Independence assumption.** The model treats all five variables as mutually independent during sampling. In reality, climate variables are correlated — for instance, high temperatures tend to be associated with reduced cloud cover. Ignoring these correlations may underestimate the probability of certain extreme combinations. Future work should explore paired or vine-copula-based LHS extensions that respect known inter-variable dependencies.
- **Sensitivity to distributional assumptions.** The output distribution depends heavily on the choice of marginal distributions and their parameters. Where these are based on assumed values rather than observational data, mis-specification can propagate into biased risk estimates.
- **Simplified physical model.** The three-stage melt chain linearises several non-linear radiative and thermodynamic processes. Albedo feedback, re-freezing, and orographic effects are not captured, limiting the physical fidelity of the simulation.
- **Static parameter values.** All constants are held fixed throughout the simulation, which does not reflect temporal variation in the glacier’s physical properties over a melt season.

Final Conclusion

The transition from the deterministic baseline through Monte Carlo to Latin Hypercube Sampling represents a principled and measurable improvement in the system’s ability to model glacier melt under uncertainty. LHS delivers lower variance, faster convergence, and more reliable extreme-risk characterisation while preserving the physical integrity of the melt model. The challenges identified — particularly the independence assumption and simplified physics — provide a clear roadmap for future milestones: incorporating correlated sampling, refining distribution parameters from observational data, and extending the model toward spatially explicit, non-linear representations of glacier dynamics.

End of Submission